

Difficulty Analysis of Posiform-Planted QUBO Benchmarks via Energy and Hamming Metrics

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Abstract—Benchmarking QUBO solvers requires instances whose optimum is known yet that remain hard. Posiform planting provides such instances with a unique planted optimum; a recent variant adds random spin-glass terms and scales the planted term by a coefficient α that tunes hardness. Hardness, however, is graded by the success rate, which is identically 0% across the hard regime and so cannot rank instances, solvers, or α values within it. We instead measure, over failing runs only and on a fixed set of the hardest instances, the energy gap to the optimum and the Hamming distance (HD) from the planted answer; together they turn the flat 0% into a continuous difficulty curve. Under simulated annealing on a D-Wave Pegasus graph ($n=5612$) the curve is non-monotone in α – a peak, then a decline – with the two quantities opposed: as α grows the failing samples move closer in HD yet deeper in energy. A single supporting run on a D-Wave Advantage annealer reproduces the same qualitative shape on much larger gaps, consistent with the structure not being specific to SA. Reporting both failure-conditioned quantities enables finer-grained grading of planted QUBO benchmarks in the zero-success regime.

Index Terms—QUBO benchmarking, posiform planting, quantum annealing, simulated annealing, Hamming distance, energy gap, α -difficulty.

I. INTRODUCTION

Benchmarking programmable quantum annealers and classical heuristics on QUBO problems requires instances whose optima are known yet non-trivial to find. Posiform planting [1] produces unique-ground-state QUBOs but is easy for simulated annealing (SA) [2]. Pelofske *et al.* [3] increased its hardness by adding random discrete-coefficient spin-glass QUBOs R_i to a posiform-planted QUBO P (unique minimum x^*) through a posiform scaling coefficient α :

$$Q_\alpha(x) = \sum_i R_i(x) + \alpha P(x). \quad (1)$$

This work was supported by the Korea Internet & Security Agency (KISA) grant funded by the Korean government (Ministry of Science and ICT, MSIT) through the “Information Security Workforce Development (Information Security Specialized University Program)” in 2026. This research was also supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (No. 25411243).

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That study tuned hardness mainly through the random-QUBO size and the annealing time, fixing $\alpha \in \{0.01, 0.1\}$, and reported the *ground-state success rate*. But the success rate is identically 0% across the entire small- α regime, so it cannot rank instances, solvers, or α values *within* it. Pelofske *et al.* conjecture the hardness stems from “local minima whose energy is very close to the true ground state.”

We make that conjecture quantitative. ΔE proxies local-minimum depth, but over all samples it conflates depth with the success rate; we therefore condition on failure and restrict to the 100 hardest instances, held constant as α varies. So measured, failure-conditioned ΔE and Hamming distance (HD) reveal a $\Delta R + \alpha \Delta P$ structure with a non-monotone peak. Neither quantity is new; our contribution is the protocol – failure-conditioning on a fixed hardest set – that converts the uninformative 0% rate into a rankable curve, with a single supporting quantum-annealer run consistent with the structure reflecting the instances rather than the SA implementation alone.

II. METHODOLOGY

Instances. We generate 500 posiform-planted instances on the D-Wave Pegasus P16 graph [4] ($n=5612$ qubits) with lin2 coefficients ($R_{ij} \in \{-1, +1\}$). Kernighan–Lin bisection splits the graph into ≤ 16 -qubit blocks, each R_i solved exactly. The planted target x^* – the concatenation of the block ground states – is the global minimum of $\sum_i R_i$; P is the posiform of a planted 2-SAT with unique solution x^* . For the difficulty analysis we rank the 500 instances by their total failing-read count under SA across the α -sweep and fix the hardest 100 as a constant set – selected from SA alone, before any QPU run, so the cross-solver agreement below is not circular – then run both solvers on it.

Metrics and flags. Per sample x we record $\text{HD}(x, x^*)$ [5] and the energy gap $\Delta E = E(x) - E(x^*) \geq 0$ under the same Q_α . With lin2 every R energy is integer, so we report ΔE in units of the minimal non-degenerate R gap ($\Delta_R=1$), comparable across instances, as is HD/n . At $\alpha=0$, x^* is one of many R ground states, so an energy match ($|\Delta E| < 10^{-7}$)

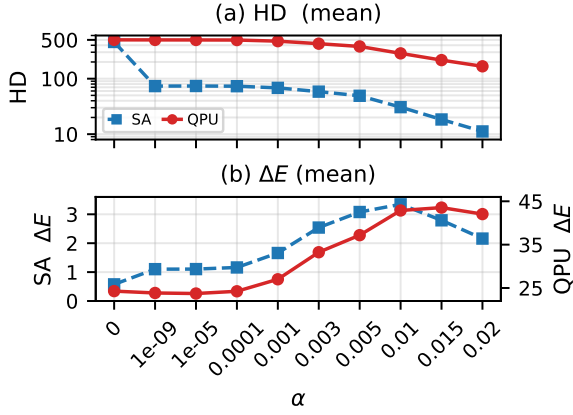


Fig. 1. Failure-conditioned difficulty over the 100 hardest instances (each α : $\geq 91\%$ of reads fail). Blue: SA; red: a D-Wave Advantage annealer. (a) HD (log axis) descends monotonically but never reaches 0. (b) ΔE on split axes (SA left, QPU right): a flat Δ_R floor (≈ 1.1), then a non-monotone peak (3.35) near $\alpha=10^{-2}$; the annealer shows the same shape on an order-of-magnitude larger baseline.

need not be a bit match; for $\alpha > 0$ it is unique. Since correct samples ($\Delta E \approx 0$, $HD = 0$) drive both quantities to 0 at large α , all statistics below are *conditional on failure* ($x \neq x^*$). Means are over failing reads; the instance-level standard error stays below $0.06 \Delta_R$ (error bars omitted).

Protocol. We sweep α over 0, 10^{-9} , 10^{-5} , 10^{-4} , 10^{-3} , 3×10^{-3} , 5×10^{-3} , 10^{-2} , 1.5×10^{-2} , 2×10^{-2} , refined near the peak; beyond it the instances are increasingly solved, leaving no fixed failing set. The main sweep runs the `neal` SA package [6] at 1000 sweeps, 100 reads per (instance, α). For a cross-solver check we replay the same 100 instances on a D-Wave Advantage annealer (Advantage_system6.4, hardware-native), with auto-scaling, two-gauge averaging, and 100 reads at $20 \mu s$; ΔE is recomputed from Q_α .

III. RESULTS

A. The Δ_R floor and the αP peak

At $\alpha=0$, R is degenerate: 56.1% of samples reach an R ground state ($\Delta E=0$) yet 0% equal x^* , with HD maximal (459, 8.2% of n); ΔE alone would read as solved, while only HD shows no sample is near x^* . For $\alpha > 0$ (Fig. 1b), in the small- α region ($\alpha \in [10^{-9}, 10^{-4}]$) ΔE stays flat at the bare R depth – mean 1.10, median exactly 1 – because $\alpha \Delta P$ is negligible; HD is likewise flat (≈ 73). As α increases past 10^{-3} the planted term becomes significant and ΔE rises to a *non-monotone peak* of 3.35 at $\alpha=10^{-2}$ – $3.0 \times$ the Δ_R floor – then declines to 2.16; floor, peak, and decline separate by far more than the instance-level SE ($< 0.06 \Delta_R$). HD meanwhile decreases monotonically ($73 \rightarrow 11$): failing samples move closer to x^* in HD even as their energy gap grows – opposite trends on the same samples. The peak follows from $\Delta E = \Delta_R + \alpha \Delta P$: $\Delta_R, \Delta P$ are per-bitstring constants, so as failures move toward x^* their ΔP falls and ΔE peaks where rising α stops outpacing it. The peak is neither a survivorship nor a selection artifact: the 100 instances fail $\geq 91\%$ of reads at every α (0% ground-state success, as [3] report), and the

same non-monotone shape appears over all 500 instances (floor ≈ 1.1 , peak ≈ 3.0), the hardest set only sharpening it.

B. Quantum-annealer validation

As supporting evidence, a single run of the same 100 instances on a D-Wave Advantage annealer (Fig. 1, red) reproduces the qualitative pattern – 0% success, a non-monotone ΔE peak near $\alpha=10^{-2}$, and monotonic HD decrease ($501 \rightarrow 167$) – on baselines an order of magnitude above SA’s ($\Delta E \approx 24$, $HD \approx 500$). One run cannot establish generality but is consistent with the structure being an instance property, not SA-specific. The peak’s couplers map to Ising $J \approx 0.03$, near the device’s integrated control error (ICE); we *hypothesize* that smaller problem scales would push them below ICE and flatten the peak, leaving this to future verification.

IV. DISCUSSION AND CONCLUSION

Failure-conditioned ΔE diagnoses difficulty (local-minimum depth) and HD diagnoses proximity to x^* , so reporting both separates how *deep* a failure is from how *close* it is. The all-sample mean (the success rate) obscures the peak, falling at larger α only because more reads succeed, not because the remaining failures are shallower. ΔE also separates solvers the success rate cannot: on 150 instances at $\alpha=10^{-3}$, 1000 vs. 5000 SA sweeps (both 0% success) give ΔE 1.68 vs. 1.00. Because the curve persists across SA budgets and is echoed by the supporting QPU run, we recommend reporting failure-conditioned ΔE (floor and peak) with HD over a fixed hardest set whenever an α -like parameter is swept; this aims to characterize the benchmark instances, though absolute values remain solver- and schedule-dependent.

Limitations. The QPU evidence is a single run (two gauges, one $20 \mu s$ anneal); the classical side is a single SA family, and instances use one coefficient type (`lin2`) and one topology (Pegasus).

Future work: broader classical solvers, more annealer gauges and anneal times, other coefficient types and topologies, and joint HD- ΔE analysis. Code and instances: https://github.com/YIDEUNKIM/qubo_dataset.

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